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## MODEL SELECTION AND VALIDATION REPORT ( remove\_5\_biomass model )

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## OBSERVATIONS:1701

## SANITY: OK - all models in the selection process passed sanity checks.

## SELECTED RANDOM TERM: spatiotemporal\_random\_fields\_ar1

## MODEL SUMMARY ( remove\_5\_biomass model )

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed\_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

|                   | coef.est | coef.se |
|-------------------|----------|---------|
| (Intercept)       | 5.90     | 0.34    |
| remove_5_kde_fric | -0.12    | 0.03    |
| remove_5_kde_feve | -0.52    | 0.04    |
| sbt_mean          | 0.24     | 0.11    |
| sbt_sd            | -0.03    | 0.05    |
| sst_sd            | 0.00     | 0.05    |
| log_chla_mean     | 0.02     | 0.04    |
| depth             | -0.05    | 0.05    |
| log_sum_fishing   | 0.11     | 0.03    |

Dispersion parameter: 1.05

Spatiotemporal AR1 correlation (rho): 0.95

Matérn range: 335.38

Spatiotemporal marginal AR1 SD: 1.02

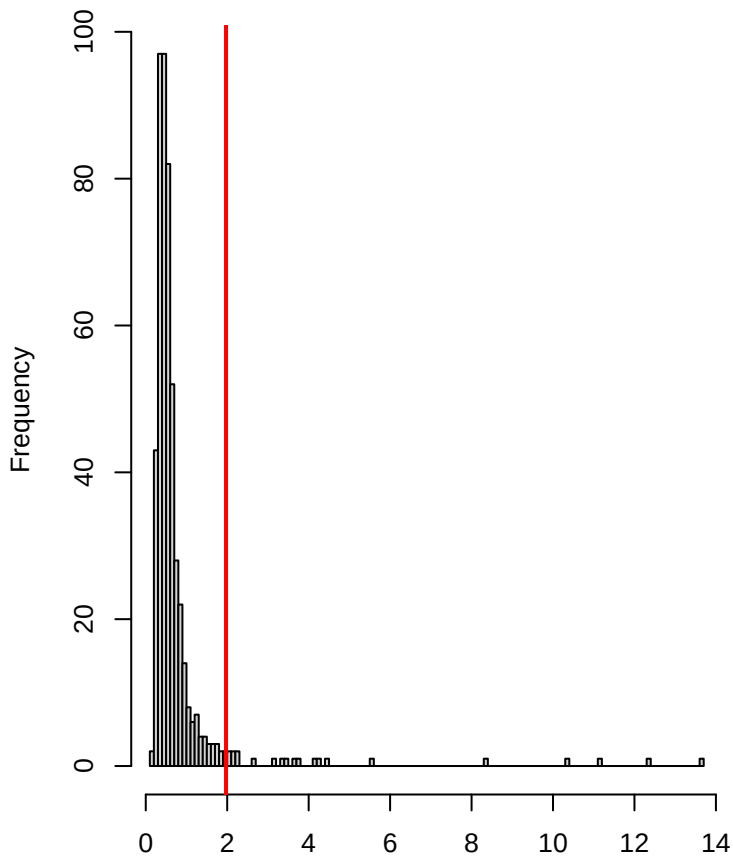
REML criterion at convergence: 11384.551

See ?tidy.sdmTMB to extract these values as a data frame.

## SELECTION STATISTICS

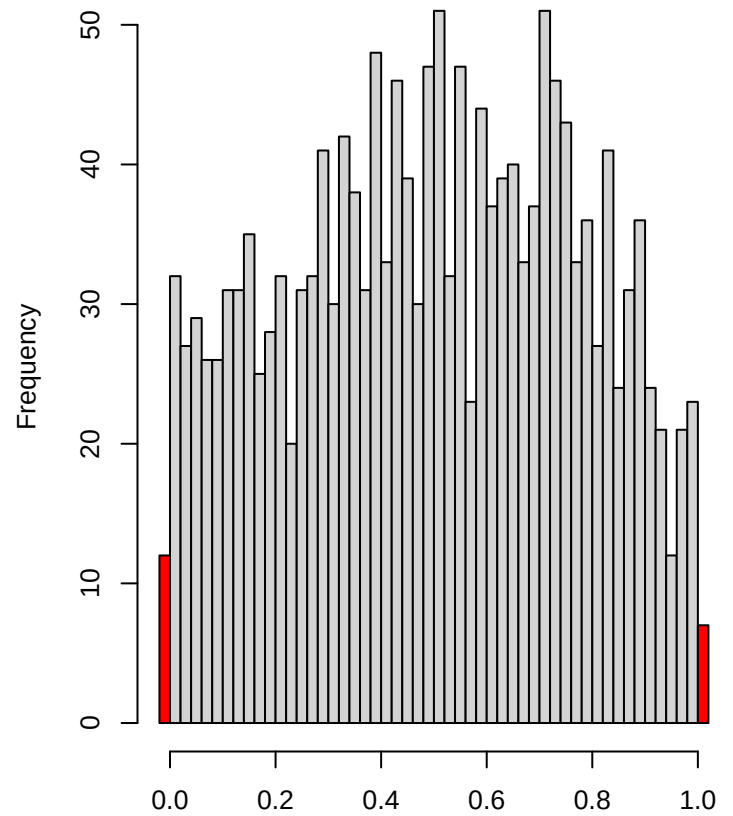
|                                                  | df | AIC      | delta_AIC   |
|--------------------------------------------------|----|----------|-------------|
| spatiotemporal_random_fields_ar1                 | 13 | 22795.10 | 2.3735637   |
| spatiotemporal_random_fields_ar1_with_anisotropy | 15 | 22797.47 | 147.5550606 |
| spatial_random_fields                            | 12 | 22945.03 | 3.3942825   |
| spatial_random_fields_with_anisotropy            | 14 | 22948.42 | 21.5625827  |
| spatiotemporal_random_fields_iid                 | 12 | 22969.99 | 0.6917749   |
| spatiotemporal_random_fields_iid_with_anisotropy | 14 | 22970.68 | 238.0336840 |
| error_only                                       | 10 | 23208.71 | NA          |

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



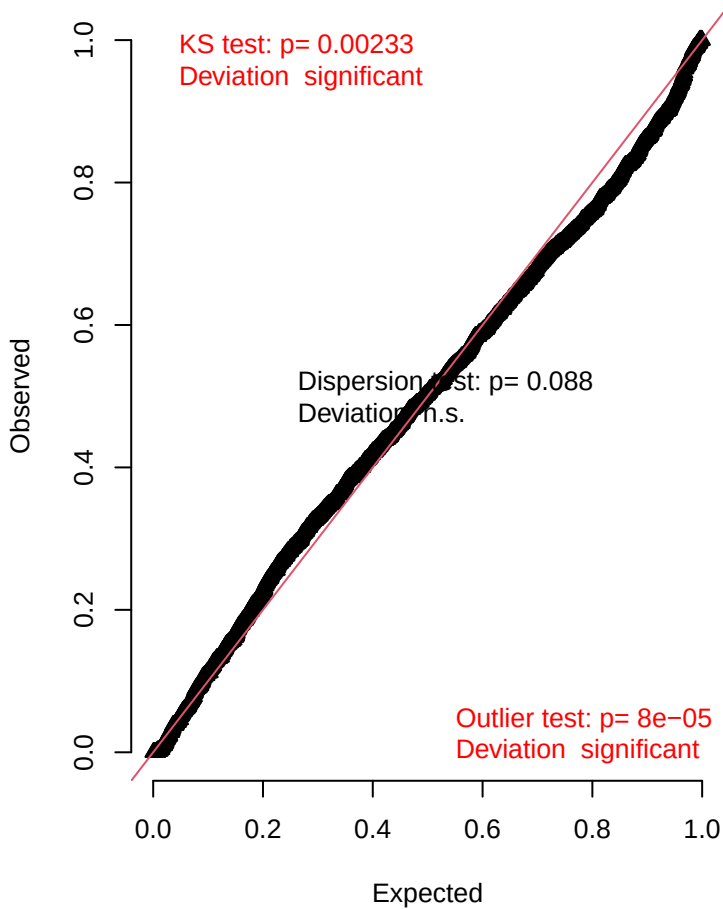
Simulated values, red line = fitted model. p-value (two.sided) =

Outlier test significant

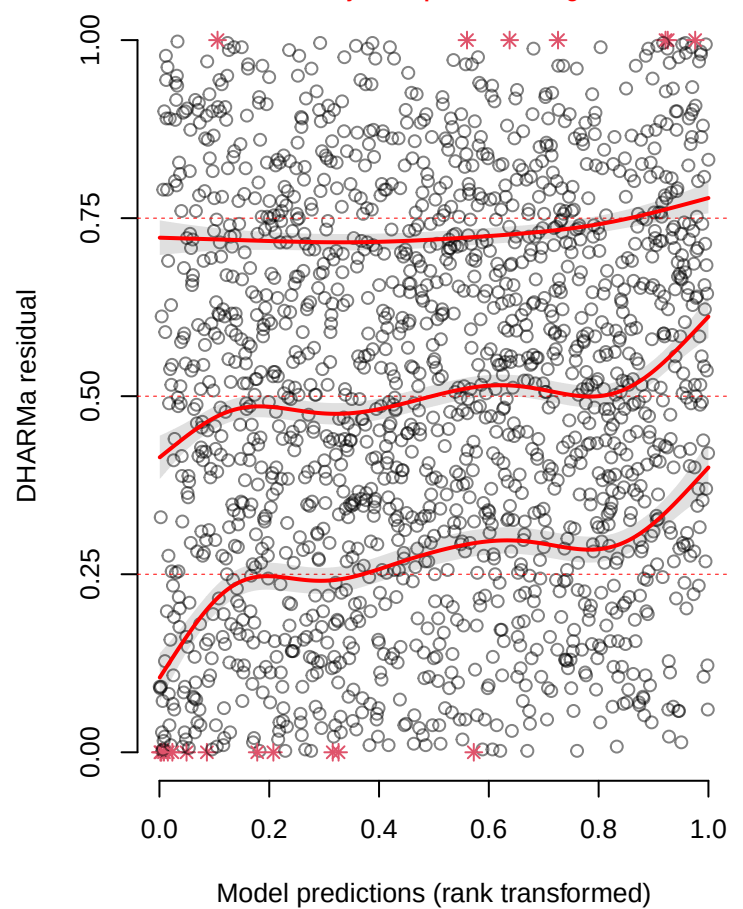


Residuals (outliers are marked red)

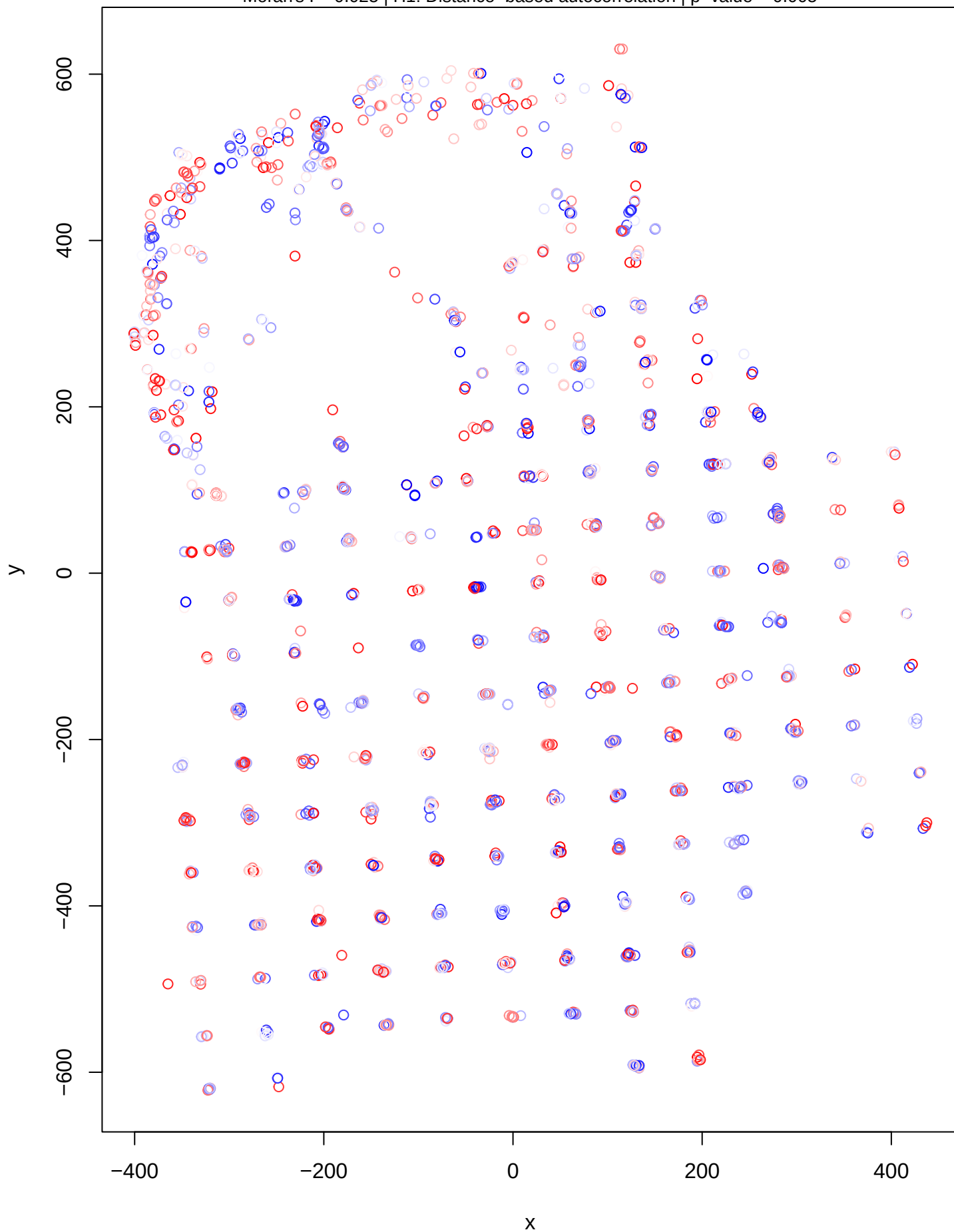
QQ plot residuals



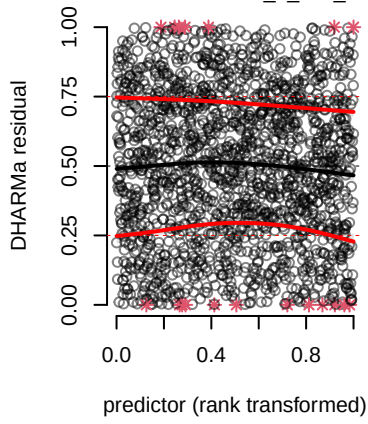
DHARMa residual vs. predicted  
Quantile deviations detected (red curves)  
Combined adjusted quantile test significant



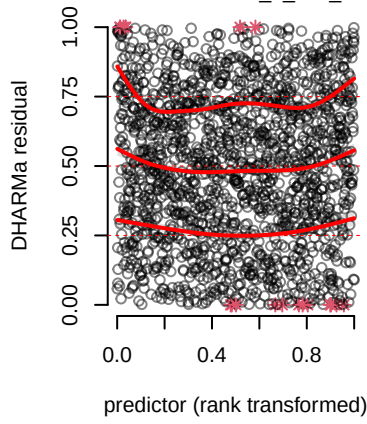
**DHARMA Moran's I test for distance-based autocorrelation**  
Moran's I = 0.025 | H1: Distance-based autocorrelation | p-value = 0.005



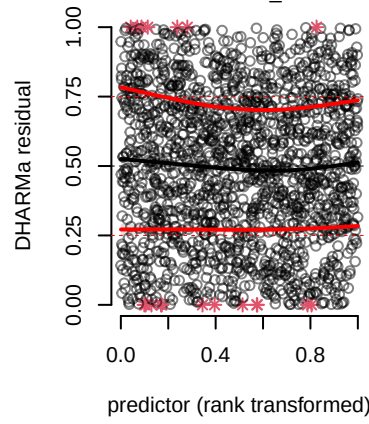
DHARMA residual Residual vs. predictor  
Quantile deviations detected (red curves)  
Combined adjusted quantile test significant



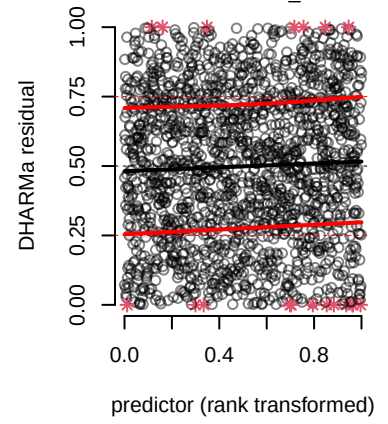
DHARMA residual Residual vs. predictor  
Quantile deviations detected (red curves)  
Combined adjusted quantile test significant



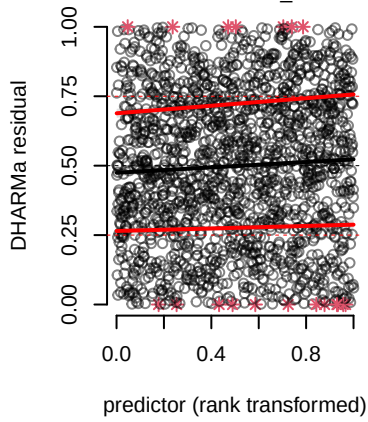
DHARMA residual Residual vs. predictor  
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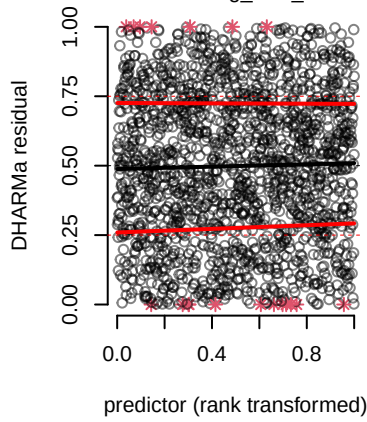
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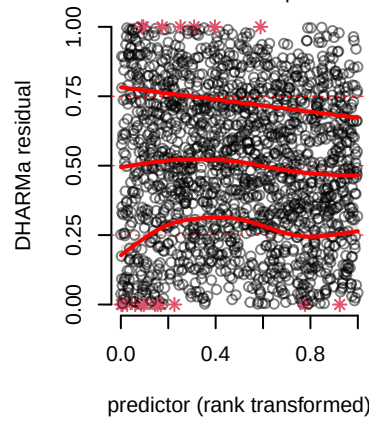
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